

Worksheet - 1

Ques 1 Why is HTML case insensitive and XML case-sensitive? Reason your answers.

Solⁿ. HTML is case insensitive

HTML was designed as a markup language for the web, mainly for displaying documents.

~~The early web was~~ Because the web needed to grow quickly. If HTML had strict rules early on, websites would break often. So browsers became forgiving.

~~Also HTML parsing is standardized today by~~

- XML is case sensitive, XML was designed very differently. XML is meant for Data exchange, structured information, etc. XML is case-sensitive because XML is used like a data format, not a display language. If XML ignored case, it could cause ambiguity also XML must support: Strict schema validation, Unique element names, so case sensitivity is necessary.

Ques 2 If HTML/CSS/JS can build websites with validation and logic, why is server side programming needed?

Solⁿ Even though HTML/CSS/JS can create interactive websites, client side code is not enough for real applications:

- (A) Security → Client-side JS runs in the user's browser. So user can view the code, modify the code and bypass validations. So server must validate.
- (B) Database access → Browser can't safely connect to the database directly eg: login system, student records,

payments. If browser had DB credentials so anyone could steal them. So server handles queries, authentication and updates.

- (C) Authentication and Authorization : To control access like who is logged?, what's their role? etc. This requires server-side sessions, tokens, cookies, etc.
- (D) File Handling : Uploading files (pdf, images, assignments) requires server-side code. Browser can't store files permanently for everyone.
- (E) APIs and Integration : Server is needed to interact with payment gateway, email server, cloud storage, and external APIs.

Ques 3 List 3 Server-Client protocols and the chief applications they are used for.

- Solⁿ
1. HTTP/HTTPS : used for web browsing, websites and web apps. Eg, Chrome requests a webpage from a server. HTTPS is more secured version.
 2. FTP/SFTP : used for file transfer between client and server. Here, FTP is older and not secure. SFTP is. FTP stands for File Transfer Protocol and SFTP for Secure File Transfer Protocol.
 3. SSH : used for secure remote login and command execution between client and server. SSH stands for Secure Shell.

Ques 4 Explain the difference between cookies and cache in http.

Solⁿ • Cookies are small pieces of data stored by browser, sent back to server in every hosted.

cookies are used for login sessions, remembering user, personalization and tracking.

- Cache is stored copies of web resources such as images, CSS, JS and HTML pages. The purpose is faster loading, reduce server load and reduce bandwidth like browser stores style.css locally so it doesn't download again.

Ques 5 What does a HTTP 302 response code indicate?

Solⁿ It means resource requested is temporarily located at a different URL. The browser should redirect to the new location.

Ques 6 Explain the disadvantages of CGI scripting.

Solⁿ Disadvantages :

- Very slow and inefficient : For every request server starts a new process, loads interpreter, executes script and returns output which is expensive.
- Hard to maintain : Compared to modern framework CGI scripts are less modular, less reusable and messy.
- High memory and CPU usage : If 100 users request at once so 100 processes created. So CGI does not scale.

Ques 7 List 3 features of HTML that may be browser dependent - test them in Firefox, Chrome, Comet.

Solⁿ Feature : 1 HTML form validation

- Different browsers show different - validation messages, styles and strictness of validation.

• Feature: 2 File Upload

- The upload button style / UI changes in different browsers
- Some browsers allow multiple file selection differently
- Drag-drop file behavior can differ.

• Feature: 3 Date Input

- Different browser show different date-picker UI
- Some browsers may have limited support
- Format and calendar behavior can vary.

Ques 8. Given this schema, find error with the XML element following it.
Correct schema,

Solⁿ.

```
<xs:complexType name="StudentType">
  <xs:sequence>
    <xs:element name="name" type="xs:string"/>
    <xs:element name="age" type="xs:integer"/>
    <xs:element name="email" type="xs:string"/>
  </xs:sequence>
</xs:complexType>
```

```
<student>
```

```
<name> 345612 </name>
```

```
<age> 2 2 </age>
```

```
<email> alice@gmail.com </email>
```

```
</student>
```

Ques 9. Give a DTD for the XML element student in Question 8.
Will the XML error be validated with the DTD?

Solⁿ `<!ELEMENT student (name, age, email)>`
`<!ELEMENT name (#PCDATA)>`
`<!ELEMENT age (#PCDATA)>`
`<!ELEMENT email (#PCDATA)>`

Ques 10. XSD for book with restriction (isbn 13 digits, price > 0)

Solⁿ `<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"`

`<xs:simpleType name = "isbnType">`
`<xs:restriction`

Solⁿ `<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"`

`<xs:element name = "book">`

`<xs:complexType>`

`<xs:sequence>`

`<!-- ISBN: exactly 13 digits -->`

`<xs:element name = "isbn">`

`<xs:simpleType>`

`<xs:restriction base = "xs:string">`

`<xs:pattern value = "[0-9]{13}" />`

`</xs:restriction>`

`</xs:simpleType>`

`</xs:element>`

`<!-- Title: string -->`

`<xs:element name = "title" type = "xs:string" />`

`<!-- Price: must be > 0 -->`

`<xs:element name = "price">`

`<xs:simpleType>`

<xs:restriction base="xs:decimal">

<xs:minExclusive value="0"/>

</xs:restriction>

</xs:simpleType>

</xs:element>

</xs:sequence>

</xs:complexType>

</xs:element>

</xs:schema>